Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam D · Form 2: Multi-step reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A 2 kg cart on a frictionless horizontal plane experiences 12 N east and 11 N north. Find the acceleration magnitude.

(A) 8.139 m/s²

(B) -8.139 m/s²

(C) 16.279 m/s²

(D) 4.07 m/s²

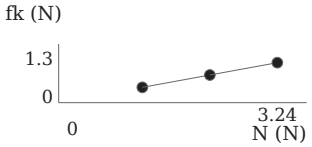
2. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. Switch to a surface with half the kinetic coefficient at the same normal loads. What is the new ratio of vertical change to the same horizontal change?

(A) 0.2

(B) 0.284

(C) 0.032

(D) 0.452



3. A 10 kg box slides right. A 70 N force acts rightward at 20° above horizontal. $\mu_k = 0.15$; contact is maintained and vertical acceleration is zero. Find normal force. Use g = 9.8 m/s².

(A) 74.059 N

(B) -74.059 N

(C) 148.117 N

(D) 37.029 N

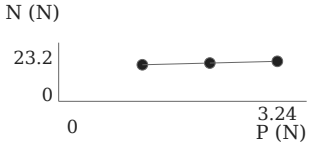
4. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1.24, 21.24); (2.24, 22.24); (3.24, 23.24). Which comparison distinguishes vertical zero bias from slope error?

(A) Force every fit through the origin.

(B) Remove units before comparing.

(C) Repeat at only one setting.

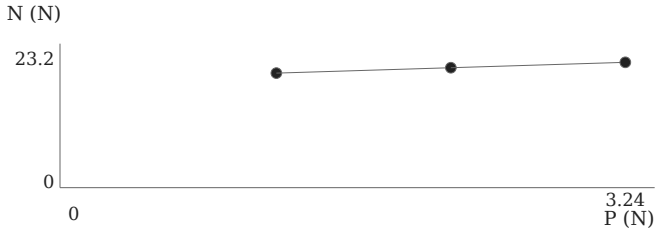
(D) Compare observed minus predicted residuals at separated settings.



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1.24, 21.24); (2.24, 22.24); (3.24, 23.24).

a. For the downward load setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Double block mass, keeping the applied P itself as the horizontal coordinate. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1.24, -0.38); (2.24, 0.12); (3.24, 0.62).

a. Build the model for cart under tow from physics and predict both slope and intercept. **(2 pt)**

b. Double cart mass while holding external resistance at 2 N. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. N (N) -> fk (N): (1.24, 0.496); (2.24, 0.896); (3.24, 1.296). Build the model for sliding-friction measurements from physics and predict both slope and intercept. **(3 pt)**